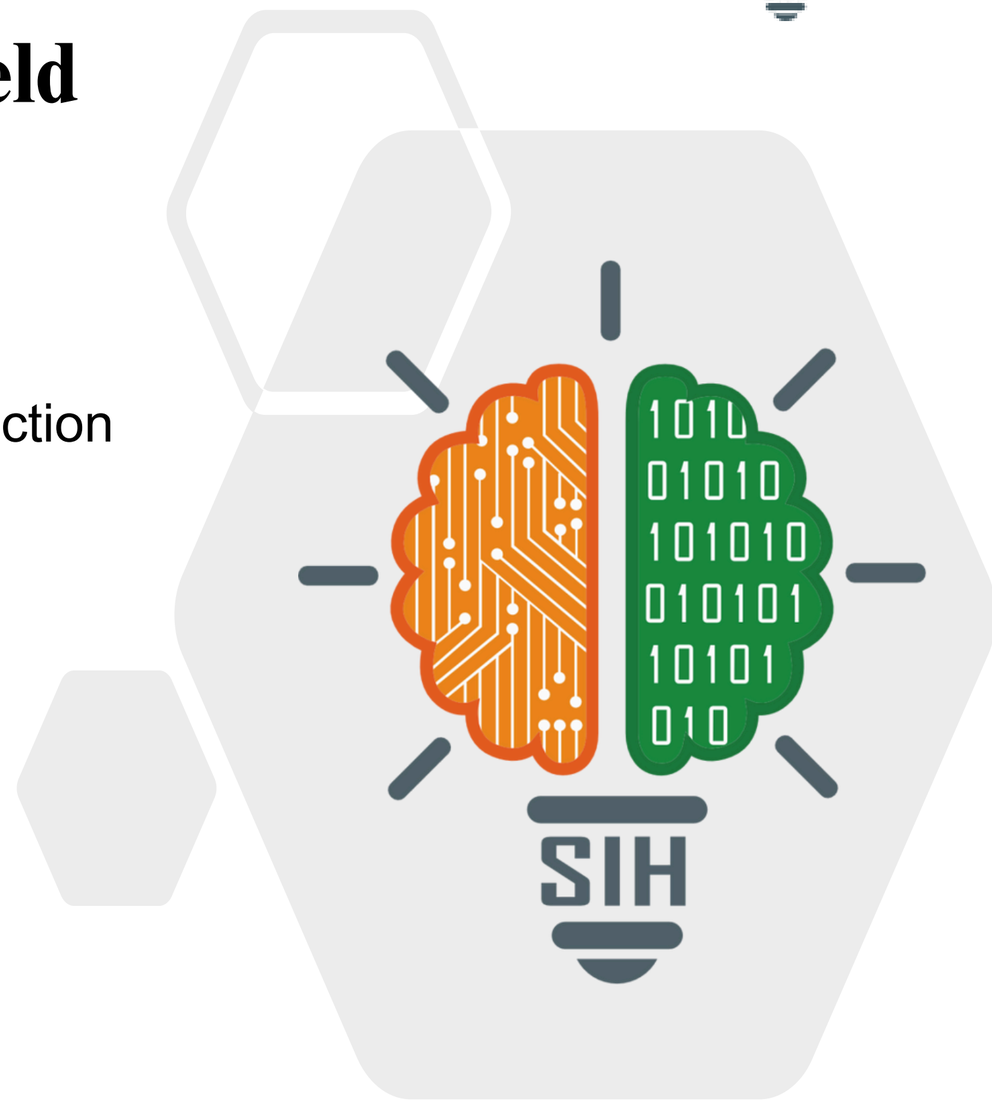
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Team- **Rock_Shield**

- **Problem Statement ID -** 25071
- **Problem Statement Title -** AI-Based Rockfall Prediction and Alert System for Open-Pit Mines
- **Theme -** Disaster Management
- **PS Category -** Software
- **Team ID -** 77901
- **Team Name -** Rock_Shield



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SMART ROCK

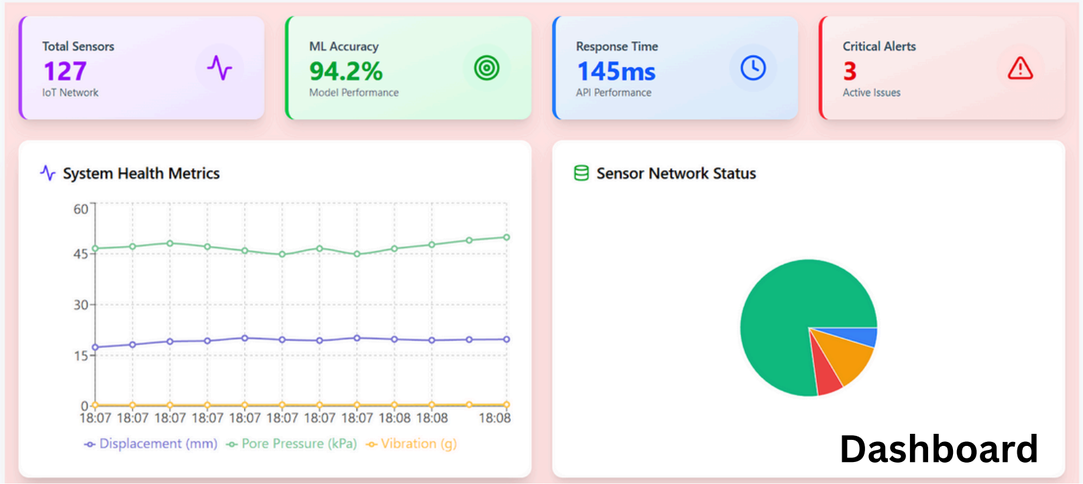
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Proposed Solution

- Proposed Solution – **SmartRock**
- **AI Prediction:** XGBoost + LSTM models forecast rockfalls in advance; explainable results.
- **Synthetic Data**: DEMs + sensor/weather simulations solve data scarcity.
- **IoT Fusion**: Low-cost sensors (tilt, vibration, pressure, weather) + GIS mapping.
- **Alerts & Dashboards:** Real-time heatmaps + SMS/email; dashboard only for site engineers to avoid confusion.
- **Ops Model**: Hybrid edge–cloud, simple setup, scalable to mines, dams, tunnels.

Unique Value Proposition

- **Predictive Safety**: From reactive detection → proactive prevention.
- **Cost-Disruptive:** Low-cost IoT sensors (15 to 20 lakhs) vs. multi-crore radar.
- **Hardware-Agnostic**: Works with any low-cost sensor setup.
- **No Data Monopoly (DEM images):** breaks the data monopoly by using a data generation engine.
- **Controlled Access**: Dashboards restricted to site engineers/managers, ensuring clear decision-misinformation among miners.



Dashboard

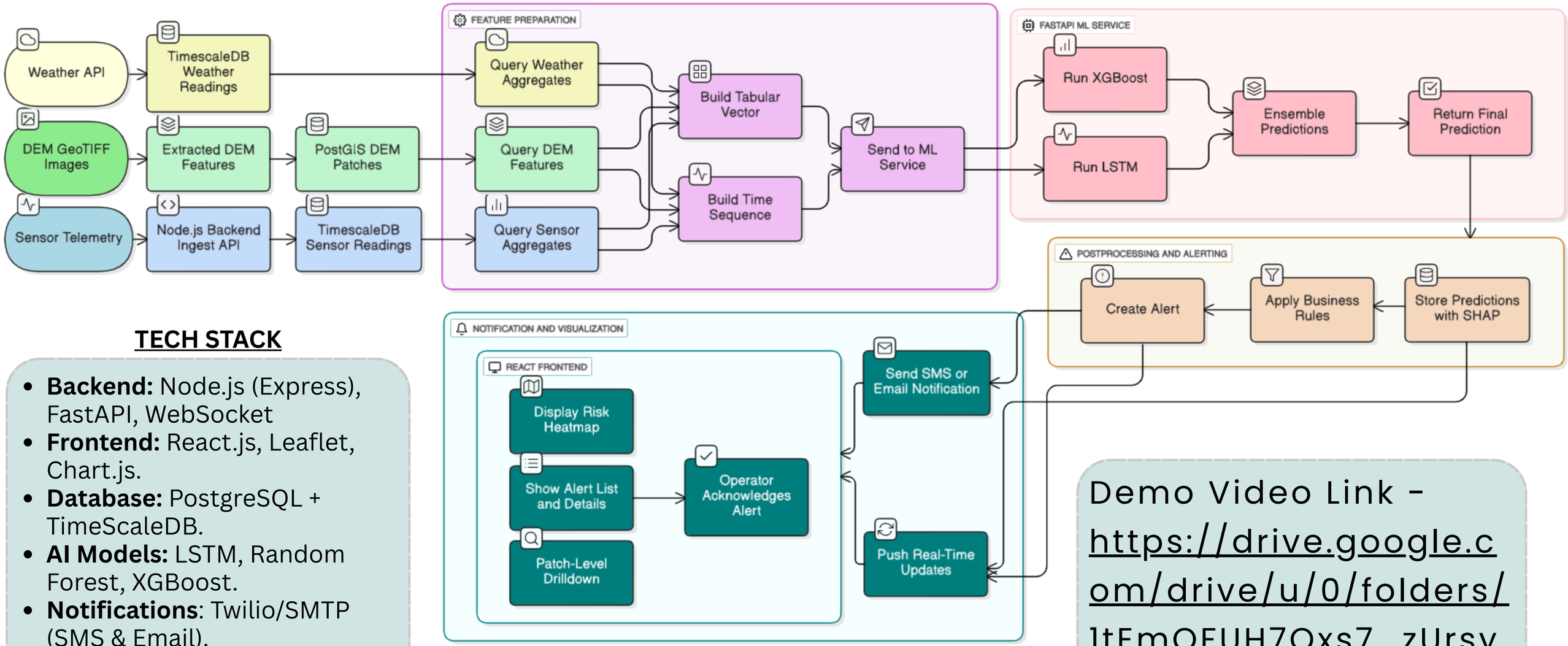
Key Characteristics

- **Relevance:** Directly addresses fatalities, equipment loss & downtime in mines.
- **Accurate Dashboards**: Heatmaps, alerts, insights.
- **Multi-Source Fusion:** DEM + IoT + weather → stronger predictions.

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TECHNICAL APPROACH

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TECH STACK

- **Backend:** Node.js (Express), FastAPI, WebSocket
- **Frontend:** React.js, Leaflet, Chart.js.
- **Database:** PostgreSQL + TimeScaleDB.
- **AI Models:** LSTM, Random Forest, XGBoost.
- **Notifications**: Twilio/SMTP (SMS & Email).
- **Preprocessing**: Python-based tools & simulations.

Demo Video Link - https://drive.google.com/drive/u/0/folders/1tEmOFUH7Qxs7_zUrsyS11DngS8FhkX2F

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FEASIBILITY AND VIABILITY

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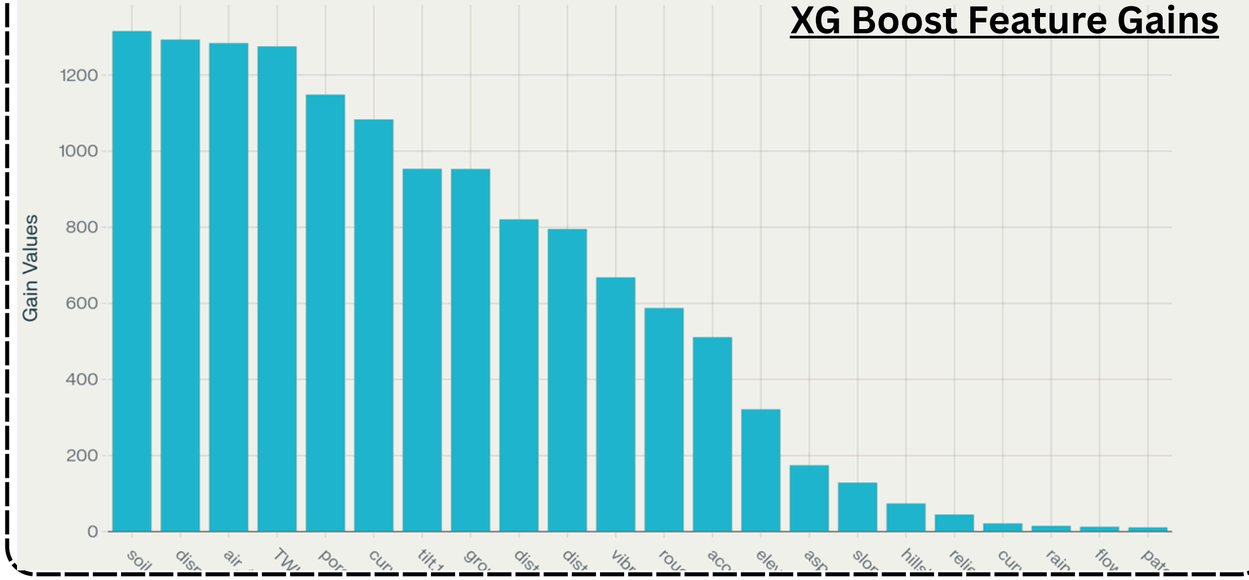
Challenges

- **Data Scarcity**: Limited labeled datasets for rockfall prediction.
- **Sensor Noise:** Commodity sensors are prone to fluctuations & false positives.
- **Connectivity Gaps:** Remote mines frequently lack stable internet connections.

Solutions

- **Synthetic Training Data:** Generate samples via DEM + weather APIs.
- **Signal Validation:** Filters, rolling averages & multi-sensor cross-checks.
- **Hybrid Architecture:** Edge processing for instant alerts + cloud sync for reliability.

XG Boost Feature Gains



Feasibillity Spectrum

Financial

Affordable for all types of Mines

Balances cost and scalability

Cost-efficient contrasting to existing systems



Technical

- Real time-capable with lightweight pipeline.
- Simple set-up.

Operational

- User-friendly dashboards.
- Flexible sensor options.

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IMPACT AND BENEFITS

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Mine Workers: Safer working conditions with early alerts

Mine Operators: Reduced downtime and higher productivity

Mining Companies: Affordable and scalable safety adoption

Regulators (DGMS, MoM): Easier compliance with digital reports

Acedemia & Researchers: Close to real world dataset for innovation

Impacts

Benefits

Social: Saves lives in mines, roads, and mountain construction zones

Economic: Low-cost, modular system that prevents losses

Environmental: Predicts failures, protects ecosystems

Scalability: Adaptable to mines, highways, tunnels

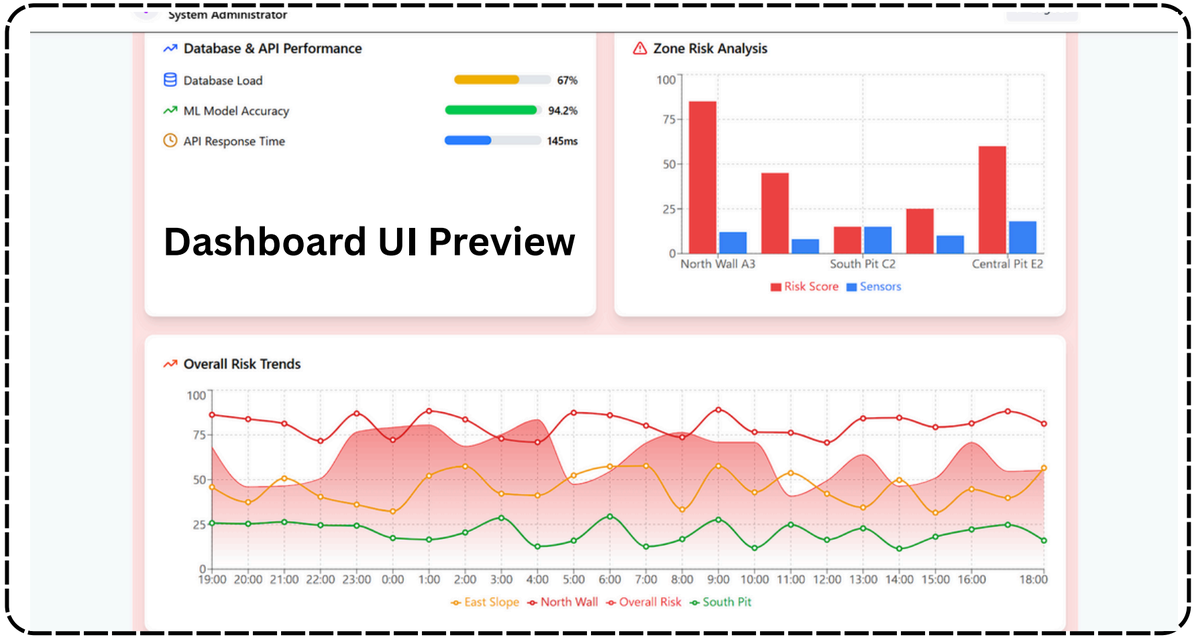
Use this QR to access:

1. Dashboard UI preview materials
2. Model Performance & feature details
3. Research summaries & references
4. DEM images
5. Demo Video
6. Dataset

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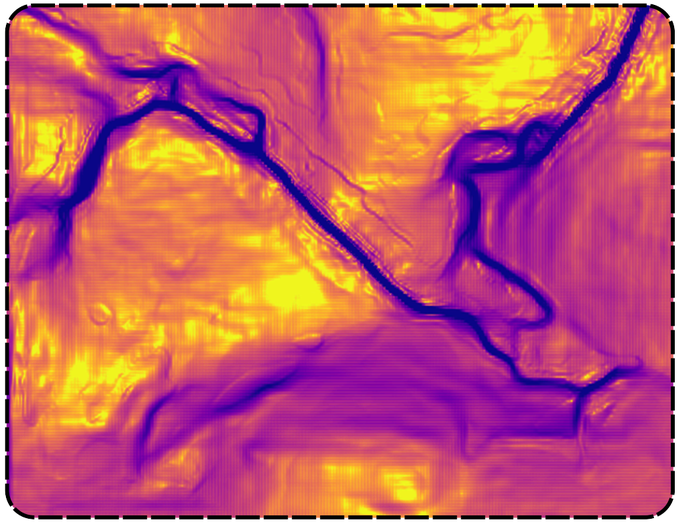
RESEARCH AND REFERENCES

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References

- Prediction of rockfall hazard in open pit mines using a ... (ScienceDirect, Senanayake 2024)
- Rockfall Dynamics Prediction Using Data-Driven Approaches: A Lab-Scale Study (MDPI — Geotechnics, Ghahramanieisalou & Sattarvand)
- Rockfall detection using LiDAR and deep learning (Farmakis et al., Engineering Geology 2022)
- Machine learning prediction of landslide deformation behaviour using acoustic emission and rainfall measurements (Engineering Geology)



Digital Elevation Model (DEM) showing terrain elevation, with brighter areas as higher ground and darker areas as valleys and drainage paths. This DEM is used to calculate the Topographic Wetness Index (TWI) for soil moisture and water accumulation analysis.

Research

Our research examines how different geotechnical sensors—such as tilt, displacement, pore pressure, vibration, soil moisture, and rainfall—capture the physical precursors of slope failure and how these signals can be combined to estimate the probability of rockfall. By analyzing their influence on slope stability and integrating them into a probabilistic framework, we translate raw sensor readings into actionable risk forecasts and show that on our dashboard.